

Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

Unit 3 assessment

Unit summary

The operating system manages resources (dividing the processor by time sharing and guarding memory), provides the user interface, hosts applications, and keeps files and devices in order; its types - batch, time-sharing, real-time, distributed, mobile - tune the same functions to different promises. File management rests on the three identifiers of a file (name and extension, folder path, metadata) and the OS's services over them. Process management rests on the process, a program in execution with its address space, registers and open files, moving through the states running, ready and blocked while the OS time-shares one processor among many. Input/output management is a translation chain: application, operating system, driver, device, smoothed by buffers and spooling - and it doubles as a diagnostic ladder. At the centre sits the kernel, the always-running core that schedules, manages memory, controls devices and exposes the guarded doorways (system calls) through which applications must request every service.

Summative assessment

Multiple-choice questions (MCQs)

1. Which of these is NOT part of the operating system's three-part job?
(Remember) a) managing the hardware resources of the computer b) providing the user-interface components c) providing a platform for applications d) typing documents for the user
2. Time sharing means: (Understand) a) each user is given a fixed hour of computer time b) the OS switches the processor among programs so quickly

that they seem to run together c) several users share one keyboard in turns
d) the computer shares its files over the network

3. A patient monitor must record every heartbeat instantly and never wait its turn. The operating system type it resembles is: (*Apply*) a) batch b) real-time c) time-sharing d) mobile
4. The three things that identify a file are: (*Remember*) a) its name, its size and its colour b) its name and extension, its folder path, and its metadata c) its icon, its font and its folder d) its drive, its cable and its port
5. A process has just started reading from the disk and cannot continue until the read finishes. Its state is: (*Apply*) a) running b) ready c) blocked d) terminated
6. Closing a window sometimes leaves a process running in the background. This shows that: (*Understand*) a) windows and processes are unrelated things b) a process is a program in execution, which may outlive its window c) the operating system has failed d) a process is a kind of file
7. A device driver is: (*Remember*) a) a type of application software b) a small piece of system software that translates the OS's requests into one device's own commands c) a network cable standard d) the part of ROM that starts the machine
8. A USB drive works on one computer but not on another. The most likely broken link in the input/output chain is: (*Apply*) a) the device itself b) the driver on the second computer c) the application d) the file's metadata
9. Applications never touch hardware directly because: (*Understand*) a) they are too slow to talk to devices b) they must ask through system calls, and the kernel decides c) hardware physically refuses application signals d) the desktop blocks them
10. The kernel is: (*Remember*) a) the desktop picture and taskbar b) the always-running core of the operating system c) a kind of application for managing files d) another name for the whole operating system

Restricted-response questions (RRQs)

1. Explain time sharing and the "illusion of many processors" in two sentences. *(Understand)*
2. Match each described machine to an operating system type, with one clause of reason: a bank's overnight payroll run; a school office desktop; a car's engine controller. *(Apply)*
3. Name the three identifiers of a file, and propose a naming pattern for a school's letters that would make them findable in five seconds. *(Understand)*
4. Narrate the states a process passes through from "the user double-clicks the icon" to "the program is closed", using the words scheduled and blocked at least once. *(Analyze)*
5. Trace a print request through the four links of the input/output chain, naming each link. *(Understand)*
6. A printer has power and paper, nothing prints, and only one computer is affected. Analyse which link of the chain the fault most likely belongs to, and name your next two checks. *(Analyze)*
7. Explain the difference between the kernel and the operating system, using the Linux example. *(Understand)*
8. Sort these into the four layers (applications, shell and utilities, kernel, hardware): the scheduler, a word processor, the desktop icons, a printer driver, the settings screen, memory allocation. *(Apply)*
9. A school computer slows down every afternoon. Explain the slowness in terms of process management and name one thing to check. *(Analyze)*
10. Explain what a buffer does and why printing does not freeze the word processor. *(Understand)*

Extended-response questions (ERQs)

1. A school wants a machine to run its biometric attendance system, which must record every check instantly. Evaluate which operating system type the machine should resemble and why, and what would go wrong with an ordinary time-sharing desktop instead. *(Evaluate)*
2. Write a one-page "office rescue plan" for a school that cannot find last term's letters and whose computer slows every afternoon: address the finding problem with a file scheme and the slowness with process management. *(Analyze)*

3. A teacher reports: "I clicked Print, the printer has power and paper, but nothing comes out, and only my computer has this problem." In one page, analyse the fault using the input/output chain, name the link you would check first, and give the likely fix at each step. (*Analyze*)
4. For a staff meeting, prepare a one-page explainer titled "Who is really running the computer?" that uses the four layers and the kernel's functions to explain why closing a window is not always ending a program. (*Analyze*)
5. Using the whole unit, evaluate the statement: "The operating system is the quiet manager of the machine." Draw on scheduling, file management, input/output management and the kernel's system calls as your evidence. (*Evaluate*)

Answers and marking guidance

MCQ answer key

1. d - doing the user's tasks is application software's job, not the OS's.
2. b - time sharing switches the processor so fast that every program seems continuous.
3. b - a real-time system must respond within a strict time limit; that is its promise.
4. b - name and extension, folder path and metadata are the three identifiers.
5. c - waiting on an operation to finish is the blocked state.
6. b - a process is the running program; the window is only one face it may show.
7. b - the driver translates for one specific device.
8. b - the device works elsewhere, so the device is fine; the missing or broken driver on the second machine is the likely link.
9. b - system calls are the guarded doorways; the kernel decides and acts.
10. b - the kernel is the core within the whole operating system package.

RRQ model answers and mark schemes

1. Model: one processor runs one process at an instant; by running one, pausing it and running another, the OS makes many programs appear

simultaneous - the illusion of many processors where there is one. **Marks (2):** 1 for the switching; 1 for the illusion.

2. Model: overnight payroll is batch (jobs collected and run without user interaction); the school desktop is time-sharing (many programs share the processor in turns); the engine controller is real-time (must respond within a strict limit). **Marks (3):** 1 per correct match with a reason.
3. Model: name and extension, folder path, metadata; a date-first pattern such as "2026-03-15-parent-notice" inside a year-then-kind folder tree makes any letter findable quickly. **Marks (3):** 1 for the three identifiers; 2 for a workable naming pattern.
4. Model: double-click creates a new process; it becomes ready; the OS schedules it into running; when it opens a file from disk it blocks until the read completes, then returns to ready and is scheduled again; on close it terminates (or keeps running in the background). **Marks (4):** 1 for new-to-ready; 1 for scheduled into running; 1 for blocked with a cause; 1 for the ending.
5. Model: the application asks the OS; the OS queues and polices the request; the driver translates it into the printer's commands; the device prints and reports back. **Marks (4):** 1 per link in order.
6. Model: since only one computer is affected, the fault sits on that machine's side of the chain - most likely the driver or the application's request, not the printer; checks: does the OS see the printer in its devices list (driver), and does the print queue show the job stuck (OS/application). **Marks (3):** 1 for placing the fault; 2 for two sensible checks.
7. Model: the operating system is the whole package - kernel plus shell plus utilities; the kernel is the essential core within it; "Linux" strictly names a kernel, and the familiar Linux desktops bundle it with shells and utilities to make complete operating systems. **Marks (3):** 1 for the package; 1 for the core; 1 for the Linux example.
8. Model: applications - word processor; shell and utilities - desktop icons, settings screen; kernel - scheduler, memory allocation; hardware - (the printer driver sits with the kernel's device-control layer under its supervision). **Marks (4):** 1 per layer mostly correct, with the driver's placement reasoned.
9. Model: each afternoon more processes compete for turns, so each gets a smaller slice and everything crawls; check the process list for background

programs or a stuck process and close what is not needed. **Marks (3):** 1 for the crowding explanation; 1 for the slice idea; 1 for a concrete check.

10. Model: a buffer is a holding area in memory where the OS parks data in transit; the word processor hands the page to the buffer and moves on while the slower printer works through it, so the application does not freeze.

Marks (3): 1 for the definition; 1 for the speed difference; 1 for the consequence.

ERQ rubrics

1. **Attendance machine (Evaluate).** Criteria: *Correct OS type with its defining promise (0-3); Recommendation justified against the instant-response requirement (0-4, Analyze); Failure mode of the time-sharing alternative explained concretely (0-2); Clarity (0-1).* Total 10.
2. **Office rescue plan (Analyze).** Criteria: *File scheme workable: at most three levels, naming pattern, rule for new files (0-3); Slowness diagnosed with process states and time sharing (0-4, Analyze); Both problems tied to OS functions (0-2); One-page clarity (0-1).* Total 10.
3. **Print fault (Analyze).** Criteria: *The chain named and applied to the symptom (0-4); Checks ordered with reasons (0-3); Likely fixes matched to the right link (0-2); Clarity (0-1).* Total 10.
4. **Staff-meeting explainer (Analyze).** Criteria: *Four layers presented accurately (0-3); Kernel functions in plain language (0-3); The window-versus-process question answered with system calls and scheduling rather than assertion (0-3, Analyze); Fits the audience (0-1).* Total 10.
5. **Integrative evaluation (Evaluate).** Criteria: *Draws on at least three of scheduling, file management, I/O management, system calls (0-3); Each drawn part used accurately (0-3); The "quiet manager" judgement defended as a synthesis, not a list (0-3, Analyze); Clarity (0-1).* Total 10.

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